



TECHNOLOGICAL AND APPLIED STUDIES FACULTY

Assessment Task Notification

Course: Enterprise Computing	
Task Type: Enterprise Project - Phase 1	Task Number: 2
Topic: Data Science + Intelligent Systems	
Weighting: 20%	Marks: 65
Issue date: Term 1 Week 6 2026	Due date: Term 1 Week 10 2026
Outcomes Assessed: EC-12-03: Explains and evaluates how data is safely and securely collected, stored and manipulated EC-12-05: Applies tools and resources to analyse complex datasets EC-12-07: explains the social, ethical and legal implications of the application of enterprise computing systems on the individual, society and the environment EC-12-08: justifies the selection and use of tools and resources to design and develop an enterprise computing system EC-12-09: selects and applies methods to record the management and evaluate the development of an enterprise computing system EC-12-11: communicates an enterprise computing solution to a specific audience	
Task Description: Overview: In phase 1 of the Enterprise project, students are required to design, develop and evaluate a Decision Support System (DSS) that addresses a clearly defined enterprise problem. The system must accept user input, process data using a knowledge base and generate meaningful decisions using an inference engine. This task integrates concepts from Intelligent Systems and Data Science units. The project is completed across four structured stages, outlined below: Stage 1: Identifying and Defining In this stage, students establish the foundation of their DSS by clearly analysing and evaluating the enterprise problem they intend to solve, clearly justifying why it requires a DSS solution. Students must describe the overall purpose of their DSS, analyse the needs of their target users	



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and evaluate the suitability of the decisions or recommendations the system will produce.

Using a top-level IPSO diagram, students illustrate how they expect the system to function at a conceptual level using the proposed knowledge base, rule base, inference process and user inputs through user interface.

At this stage, students are describing their intended design rather than implementing the system.

Students also analyse and evaluate the types of data to be used and evaluate relevant social, ethical and legal considerations related to data collection, data processing and system design.

Stage 2: Researching and Planning

In this stage, students research similar enterprise systems and analyse their strengths and limitations, with particular reference to how these systems process user inputs, the types of data they employ and the decisions they generate. Students analyse and synthesise key insights from this research and justify how these insights inform the planning of their own DSS.

Students then describe and justify the type of decision-making and inference strategy of their DSS. They must explain how the chosen inference approach influences the overall system design.

Students are required to perform a thorough data analysis and justify their selection of data type and data sources. Using this analysis, students design the knowledge base by specifying the facts and variables to be stored, as well as the structure of the stored data.

Students develop a structured rule base using clearly written IF-THEN logic and create a decision tree to demonstrate how conclusions will be reached. Where appropriate, certainty factors may be applied.

Students communicate the planned logical processes of their DSS using appropriate diagrams such as Data Flow Diagrams (DFDs), flowcharts, and decision trees.

Students design a data sourcing, collection and storage strategy that will guide their database implementation. They must plan their database structure and justify the choice of database type, data dictionary, table structure and use of primary and foreign keys. Students must also design the rule base and identify the queries required to generate DSS outputs.

Students apply UX/UI principles to design and justify their choice of user interface elements, input methods and any required hardware. They must also design a suitable output dashboard to display the decisions or recommendations generated by their DSS.



Stage 3: Producing and Implementing

In this stage, students implement their planned system. They collect or generate data and demonstrate appropriate data preparation processes, including data cleaning, validation, filtering, grouping and addressing inaccuracies or redundancies. Students should demonstrate the process of database normalisation, building a database schema, data dictionary and calculating the database size.

Students implement their knowledge base within a database using appropriate software. This includes creating tables, defining keys, designing forms to support user interactions and generating reports to display DSS outputs.

Students then implement and evaluate their rule base using queries or programmed logic to demonstrate automated decision making and generate appropriate reports showing the outcomes/decisions generated by their DSS.

Stage 4: Testing and Evaluating

In the final stage, students verify and validate their DSS using structured test cases that demonstrate automated processing of user inputs to generate outcomes. The testing must be clearly documented, demonstrated and recorded for your client with explicit reference to test cases, triggers, conditions, and actions.

Students then evaluate their DSS in terms of operability, accuracy, efficiency, ease of use, accessibility and inclusivity. Finally, students propose realistic improvements based on testing and evaluation.

Deliverables:

Students must submit:

- A DSS portfolio containing appropriate documentation for each project stage
- Your DSS attached as an Excel workbook Or a database file Or code files (with instructions on how to run it added in an Appendix in your documentation)
- A video recording for the client that introduces the system, explains its features and demonstrates its use through a comprehensive testing process

Permissible Use of A.I.:

A.I tools may be used for research purposes and for generating suitable datasets for the DSS. Any use of AI must be appropriately referenced.



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Students are required to include:

- All AI prompts used
- Clear screenshots of generated outputs linked to each prompt

They must be presented in an appendix at the end of the project document.

See A.I policy in the Stage 6 Assessment Handbook for further details.

Marking Rubric:

Stage 1 : Identifying and Defining (10 Marks)

Core descriptor:

The student **evaluates** the enterprise problem, defines the purpose of the DSS, and **analyses** the needs of target users and the decisions or recommendations to be generated. The student **explains** the intended system functionality using an accurate IPSO diagram. The student **explains** the proposed knowledge base, rule base, inference process and user interface. The student **analyses** the types of data to be used and **evaluates** relevant social, ethical and legal considerations related to data collection, processing and system design.

Level	Descriptor
Outstanding (9 - 10)	The student evaluates , analyses and explains all aspects of the DSS problem, system purpose, users, IPSO diagram, system components, data types and social, ethical and legal considerations.
High (7 - 8)	The student analyses and explains the DSS problem, system purpose, users, IPSO diagram, system components, data types and social, ethical and legal considerations.
Sound (5 - 6)	The student explains and describes the DSS problem, system purpose, users, IPSO diagram, system components, data types and social, ethical and legal considerations.
Basic (3 - 4)	The student describes the DSS problem, system purpose, users, IPSO diagram, system components, data types and social, ethical and legal considerations.
Limited (1 - 2)	The student identifies aspects of the DSS problem, system purpose, users, IPSO diagram, system components, data types and social, ethical and legal considerations.



Stage 2: Researching and Planning (25 Marks)

Core descriptor:

The student **evaluates** similar enterprise systems by **analysing** their strengths and limitations and **explains** how key insights inform the DSS design. The student **evaluates** and **justifies** the decision-making type and inference strategy. The student **analyses** data types and data sources and **justifies** their selection. The student **defines** and **explains** the structure of the knowledge base. A structured rule base is **developed** using IF–THEN logic and supported by a decision tree and certainty factors where appropriate. Logical processes are **explained** using DFDs, flowcharts and decision trees. The student **evaluates** and **justifies** the data sourcing, collection and storage strategy, including database type, data dictionary, table structures and keys. The student **evaluates** and **justifies** user interface and dashboard design using UX/UI principles.

Level	Descriptor
Outstanding (23 - 25)	The student evaluates , analyses , explains and justifies all research, planning, inference, data, database, rule base, diagrams and UX/UI decisions.
High (19 - 22)	The student analyses , explains and justifies research, planning, inference, data, database, rule base, diagrams and UX/UI decisions.
Sound (14 - 18)	The student explains and describes research findings, inference choices, data selection, rule base design, diagrams, database planning and UX/UI decisions.
Basic (9 - 13)	The student describes research, inference strategy, data choices, rule base, diagrams, database planning and UX/UI decisions.
Limited (1 - 8)	The student identifies elements of research, inference strategy, data, rule base, diagrams, database planning and UX/UI design.

Stage 3: Producing and Implementing (20 Marks)

Core descriptor:

The student **evaluates** the implementation of a fully functional DSS. Data is **collected**, **processed**, **cleaned**, **validated**, **filtered** and **grouped** effectively. The student **explains** the database implementation, including tables, keys and normalisation. The student **explains** how the rule base is implemented using queries or programmed logic to demonstrate automated decision-making. Reports **clearly communicate** system outcomes.

Level	Descriptor
Outstanding (18 - 20)	The student evaluates and explains the full DSS implementation, including data preparation, database design, rule implementation and



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	automated outputs.
High (15 - 17)	The student analyses and explains the DSS implementation, including data preparation, database design and rule implementation.
Sound (11 - 14)	The student explains and describes the DSS implementation, including data preparation, database design and rule implementation.
Basic (7 - 10)	The student describes the DSS implementation, including data preparation, database design and rule implementation.
Limited (1 - 6)	The student identifies aspects of data preparation, database implementation and rule logic.

Stage 4: Testing and Evaluating (10 Marks)

Core descriptor:

The student **evaluates** the DSS using structured test cases. Testing is **explained** with reference to test cases, triggers, conditions and actions. The student **analyses** and **evaluates** system operability, accuracy, efficiency, ease of use, accessibility and inclusivity, and **justifies** realistic improvements.

Level	Descriptor
Outstanding (9 - 10)	The student evaluates , analyses and justifies testing results and system improvements.
High (7 - 8)	The student analyses and explains testing results and identifies improvements.
Sound (5 - 6)	The student explains and describes testing results and system effectiveness.
Basic (3 - 4)	The student describes testing results with limited evaluation.
Limited (1 - 2)	The student identifies limited testing evidence or outcomes.

IMPORTANT NOTE: If you are absent on the day of the task, you must follow School [illness/misadventure application](#) procedures and see the Head Teacher on your next day of School attendance. Otherwise a zero mark will be awarded and a course warning issued.



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ASSESSMENT PROCEDURES

This is a brief summary of the assessment policy at Arthur Phillip High School. Please see the assessment policy for clarity and further information. This summary does not supersede the policy.

It is the student's responsibility to be alert to the notification of tasks. In case of absences from school, students will need to check with their teachers upon their return if they have missed any notifications of upcoming tasks. Refer to the online policy for more detail.

1. Students who **fail to submit** an assessment task by the due date and time **8:45 am may be given zero** and a course warning if they have no valid reason.
2. Students who **fail to attend** an assessment task which is to be completed at School **may be given zero** and a course warning if they have no valid reason.
3. All work that is submitted for assessment purposes must be the independent work of the student concerned (or where group work is permitted, of the students concerned). Student malpractice including plagiarism, collusion, breach of assessment conditions or a non-serious attempt **may result in a zero** being awarded.
4. Students **should not miss** any timetabled periods, nor come to school **late** without a valid reason on the day an assessment task is to be completed. It may be considered malpractice, as it could be seen as gaining an unfair advantage over other students.
5. If a student submits a task which is deemed to be a **non-serious attempt** then a zero mark may be awarded. Students should complete an assessment task knowing they **must** make a genuine effort at the assessment task or examination. A student's task or exam is considered a **non-attempt** if there is no evidence of academic engagement with the task or exam. Students are required to attempt a range of question types throughout the assessment task or examination paper.
6. If you believe factors beyond your control, e.g. sickness, family misfortune, have affected your ability to do your best on a task, you may apply for an extension by seeing the appropriate Head Teacher of the course at least 24 hours before the due date. No extensions will be granted to applications received on the day a task is due.
7. Students who **fail to submit** an assessment task by the due date or **fail to attend** a task that is to be completed at school, but **who have a valid reason** (such as sickness or approved leave validated by appropriate documentation) are expected to follow Illness/Misadventure processes.
8. If a student is **absent** on the due date of an assessment task, it is the student or parent/guardian's responsibility to phone the school (**Telephone: 8855 8100**) and inform the Head Teacher of the subject of the student's absence. An illness/misadventure appeal application form should be completed and submitted **online** with accompanying evidence (C1 form and a Doctor's certificate). This must be submitted within 48 hours. Hard copies of evidence should be submitted to student services. Students should be aware that they may be expected to complete the task **on their first day back to school, NOT the first day they have their next lesson of the impacted subject**. Failure to follow this process may result in the application being rejected and a **zero mark** being awarded.
9. Students seeking disability provisions for assessment tasks are to communicate with the Learning Support Faculty and their Teacher at least two weeks prior to the date of the assessment task. The Teacher will inform students seeking disability provisions of the exact venue and details for the task.